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(1) Research Program/Project/Study Title: Zestful Infusion: Enhancing CapSU Calamansi Juice with Calamansi-Infused Tapioca Pearls	
Proponent Name:	Kris B. Roxas
Academic Rank:	Associate Professor 1
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Pontevedra
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Food and Beverage/Calamansi		(8) Discipline/Field of Specialization Food Technology Food Chemistry	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Calamansi is recognized for its nutritional richness, particularly in vitamin C, essential for immune system support (Venkatachalam et al., 2023). Additionally, its antioxidant properties contribute to overall health and well-being. By infusing calamansi into tapioca pearls, this research not only enhances flavor but also ensures a boost of essential nutrients. Other products derived from calamansi include calamansi-infused jams, sauces, and dressings, showcasing the versatility of this local citrus fruit (Lee, 2000; Quijano, 2020). These products not only provide unique culinary experiences but also support local farmers and promote sustainable agricultural practices.

This research will be conducted in consonance with the broader Calamansi Project at CapSU, Pontevedra, a proof of the university's commitment to innovation in utilizing local resources. By incorporating calamansi into tapioca pearls, CapSU can contribute to the regional agricultural economy by promoting the use of calamansi. In terms of beverage innovation, this study envisions to produce a delightful fusion of flavors and textures, aiming to elevate the sensory experience of CapSU calamansi juice. It will focus on the incorporation of calamansi-infused tapioca pearls, introducing a novel and exciting dimension to the well-known citrus drink. The infusion of these pearls not only enhances the beverage's taste profile but also creates a visually appealing and texturally satisfying refreshment. The infusion of calamansi into tapioca pearls not only imparts a unique citrusy burst but also brings forth potential health benefits associated with calamansi, known for its rich vitamin C content and antioxidant properties (World Health Organization, 2003).

This research is in line with the university's research agenda number 2, that is, to achieve inclusive food sustainability and specialized production and to promote product development and innovation for primary and secondary commodities. With this, it seeks to add significant value to the existing CapSU calamansi juice product, transforming it into a signature beverage that captivates the taste buds and offers a memorable drinking experience.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned. .)

Tapioca pearl, also known as tapioca ball, is an edible translucent sphere produced from tapioca, a starch made from the cassava root. They originated as a cheaper alternative to sago in Southeast Asian cuisine. When used as an ingredient in bubble tea, they are commonly referred to as pearls or boba. Flavor fusion has emerged as a fascinating approach to improving flavor sensations in the world of beverage innovation. This study explores the concept of Zestful Infusion, a creative experiment that aims to improve the CapSU Calamansi juice by adding tapioca pearls to it. The combination of the refreshing vibrancy of calamansi and the fun texture of tapioca offers a unique option as consumers look for more distinctive and enjoyable drink variations. This mixture utilizes infusion method can add sensory appeal, and possible nutritional advantages when compared to existing beverages in the market. The study intends to add to the growing field of beverage production by providing insights into the science and technology of creating unique and delightful drinks. It is backed by prior research on flavor combinations and sensory evaluation study by some BS Hospitality Management students of CapSU Pontevedra (Ballejo, et al., 2023).

Furthermore, the introduction of tapioca pearls provides a layer of sophistication to the beverage, appealing to a diverse consumer base. The research aims to contribute to the product development landscape by creating an innovative and marketable refreshment that aligns with evolving consumer preferences for both flavor complexity and visual appeal. In the field of hospitality management, this product "Zestful Infusion" stands out as an exciting addition to beverage offerings. The product not only enhances the beverage options but also serves as a conversation starter, providing an opportunity for establishments to distinguish themselves in a competitive market. The unique combination of CapSU calamansi juice and calamansi-infused tapioca pearls caters to the growing demand for experiential dining, allowing hospitality businesses to elevate their guest

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experience. With this, the researchers aim to provide baseline data in terms of the physico-characteristics and acceptability of the product to guide food innovators in their formulation.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study will be conducted in order to determine the acceptability of developing the "Zestful Infusion", the incorporation of calamansi-infused tapioca pearls to CapSU Calamansi Juice.

Specifically, it has the following objectives:

1. to determine the physico-chemical characteristics of Zestful Infusion in terms of:
 - 1.1 pH
 - 1.2 total titrable acidity
 - 1.3 total soluble solids
 - 1.4 proximate, mineral and vitamin C contents
2. to determine the acceptability of Zestful Infusion in terms of:
 - 2.1 taste
 - 2.2 appearance
 - 2.3 color
 - 2.4 aroma
 - 2.5 overall acceptability
3. to determine if there are significant differences on the physico-chemical characteristics of Zestful Infusion in terms of:
 - 3.1 pH
 - 3.2 total titrable acidity
 - 3.3 total soluble solids
 - 3.4 proximate, mineral and vitamin C contents
4. to determine if there are significant differences on the acceptability of Zestful Infusion in terms of:
 - 4.1 taste
 - 4.2 appearance
 - 4.3 color
 - 4.4 aroma
 - 4.5 overall acceptability
5. to conduct cost analysis of the Zestful Infusion.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Closest Prior Art

WO2008062057A1	NATURAL EQUIVALENT OF CHEMICALLY MODIFIED STARCH
US20230371566A1	BOBA WITH IMPROVED FLAVOR AND COLOR FEATURES AND METHODS OF MAKING AND USING SAME
US20210195919A1	METHOD FOR PRODUCING PULP-CONTAINING JUICE AND PULP-CONTAINING JUICE PRODUCED THEREBY

The Calamansi Fruit

Calamansi, a citrus fruit tree also known as *Citrofortunella microcarpa* or calamondin, grows natively in the Philippines. Calamansi trees bear small, orange-like fruits, dark green foliage and white flowers. The calamansi tree grows up to about 25 feet tall at maturity, and it is hardy to U.S. Department of Agriculture zone 9 and above. Like most other citrus trees, calamansi grows readily from seed. This is a reliable propagation method if you cannot find calamansi saplings or cuttings at a plant nursery (Department of Agriculture, 2022).

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Calamansi is a smooth and slightly spiny plant, growing to a height of 3 to 5 meters. Leaflets are elliptic to oblong-elliptic, 4 to 8 centimeters long. Petioles are very narrowly or scarcely winged, about 1 centimeter long. Flowers are axillary, solitary, rarely in pairs, white, and short-stalked. Fruit is yellow when ripe, nearly spherical, 2 to 3.5 centimeters diameter, 6- to 7-celled, and thin-skinned. The skin or peel is green to yellowish green or yellow, loosely adhering to the flesh. The flesh contains a few light orange seeds. Other common names of calamansi in the Philippines are: kalamondin, kalamunding, kalamansi, limonsito and agidulce.

The calamansi tree has upright branches with thorns, and can grow up to 10 feet high. The calamansi fruit is about 1 3/4 inch in diameter, dark green when unripe and yellow orange when ripe, with loose skin, segmented contents and the pulp is very acidic. (Soyus, 2006)

Composition of Calamansi. The physicochemical and microbiological properties of calamansi highly determine its sensory characteristics, shelf-life and nutritive value. The Philippine National Standards are documents providing, for common and repeated use, rules, guidelines or characteristics for activities or their results, aimed at the achievement of the optimum degree of order in a given context.

Physicochemical Properties

Total Soluble Solids (TSS). The TSS content of the fruit is usually obtained from assessing the Brix of the fruit. The TSS or sugar content measures and includes the sucrose, fructose, and glucose, carbohydrates, organic acids, proteins, fats and minerals of the fruit. (Tiencheu, et al., 2021).

Total Titratable Acidity. The measurement of titratable acidity in fruit juice measures the concentration of titratable hydrogen ions contained in the fruit juice samples by neutralization with strong base solution to a fixed pH. This value includes all the substances of an acidic nature in the fruit juice: free hydrogen ions, organic acids, acid salts and cations (Tiencheu, et al., 2021).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

This study is anchored on the technology developed by Liu, Cheng and Wu (2021). The technology involved in crafting fruit-flavored "boba" balls holds significant potential for application by companies specializing in food ingredients. By refining the core formula of these fruit balls, it becomes feasible to incorporate health-promoting compounds such as flavones, active proteins, sterols, enzymes, probiotics, and vitamins. This advancement addresses the conventional challenges in this domain, notably enhancing the stability and compatibility of these nutritious substances when integrated into beverages. Furthermore, this innovative approach serves to effectively prolong the shelf life of these products.

In this proposed research, a less sophisticated method of infusing calamansi to boba/tapioca pearls is explored, merging it with the formulation of CapSU Calamansi juice to create a zestful infusion, a new selection of beverage to be introduced to the consumers. This is also based of the research of Ballejo et al.(2023), a group of BSHM students who developed the calamansi-infused boba pearls.

The researchers are aiming to introduce an inventive step in the preparation of Zestful Infusion, to ensure its novelty and the product's optimum quality.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Preparation of Ingredients and Materials

The ingredients will be bought from the CapSU Pontevedra's calamansi project, the local market and/or from suppliers within the vicinity. The following supplies and tools will be utilized in the study to prepare the various treatments: tapioca starch, honey, calamansi extract, and water. A strainer, mixing bowl, spatula, measuring cup, measuring spoon, measuring cup, pot, knife, cutting board, large container and butane will be among the implements and instruments to be utilized in the process.

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Experimental Design and Treatment Distribution

Four (4) treatments will be replicated three (3) times, arranged in a completely randomized design (CRD). The treatments will vary on the amount of calamansi infused in the tapioca pearls.

Treatment A:

4 tablespoon of Calamansi extract
 $\frac{1}{2}$ cup of tapioca starch
 1 tablespoon of honey

Treatment B:

6 tablespoon of Calamansi extract
 $\frac{1}{2}$ cup of tapioca starch
 1 tablespoon of honey

Treatment C:

No Calamansi Added
 $\frac{1}{2}$ cup of tapioca starch
 1 tablespoon of honey

Treatment D:

commercial tapioca pearl
 Commercially available tapioca pearl

Preparation of Tapioca Pearls and Calamansi Juice

Calamansi extract will be obtained by squeezing the calamansi fruit that has been cleansed beforehand. After sufficient juice has been extracted, the amount needed for each treatment will be determined using a measuring cup. The specified quantity of calamansi extract will be poured into a pot and subjected to medium heat. Subsequently, honey will be introduced and thoroughly blended during the initial boiling phase. Upon achieving a well-combined state, tapioca starch will be incorporated and blended. The tapioca pearl may be removed from the heat when it attains a stretchy consistency. Following this, the tapioca dough will be transferred to a sanitized surface and kneaded until it achieves an exceedingly smooth texture. After this, the dough will be shaped into multiple logs. Small squares will be cut with a knife to form uniform balls. To prevent the rolled balls from adhering to each other, an additional application of tapioca starch will be implemented. Any excess tapioca starch will be filtered out. The prepared tapioca pearls will then be placed in boiling water and heated for a duration of 15 minutes, with frequent stirring. Upon reaching a softened state, the tapioca pearls will be extracted, scooped out, and subsequently refrigerated.

Calamansi Juice Ingredients:

500 ml calamansi pure
 4 liters of water
 1 kilo of refined/brown sugar

All ingredients and equipment will be prepared in advance. Utilizing a knife and cutting board, the calamansi will be sliced, and the juice extracted to yield 500 ml of pure calamansi. Subsequently, a large container will be utilized to combine four liters of water and one kilogram of sugar. Following this, the mixture will be stirred until complete dissolution of the sugar occurs. Once the sugar has completely melted, 500 ml of pure calamansi will be poured into the mixture.

Zestful Infusion

Once the calamansi juice and calamansi-infused pearls are prepared, the plastic cups with cover and boba straw will be used in the distribution of the product. The same formulation of calamansi juice will be used for all the treatments.

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Physico-chemical Analysis

The methods to be employed in the analysis will be based on the Association of Official Analytical Chemists (AOAC) Methods. The tests to be conducted will be as follows:

- 1.1 **pH..** It will be determined in ten milliliters of the sample dispensed into a beaker after calibrating the pH meter with phosphate buffer of pH 4.0 and 7.0 (Adubofuor et al., 2010; AOAC, 1990).
- 1.2 **total titrable acidity.** Total titratable acidity (TTA) as % malic acid will be determined by titration using 0.1 N NaOH and phenolphthalein indicator (Talasila et al., 2012; AOAC, 1990).
- 1.3 **total soluble solids.** Total soluble solids will be determined by weighing an empty Whatman No. 1 filter paper and then passing a known mass of sample through it. It will then be dried inside a ventilated oven at 103°C for 2h, and the solid left on the filter after evaporation will be weighed and used to calculate the TSS, which is the material remaining on the filter after moisture have been evaporated (Adubofuor et al., 2010; AOAC, 2005).
- 1.4 **proximate, mineral and vitamin C contents.** Moisture content will be determined using the moisture content analyzer in the CapSU-SRA Laboratory. Ash Content will be determined in the same laboratory, using an accurately weighed empty crucible and then 10ml of sample will be weighed in it using an analytical balance. The sample in crucible will placed in muffle furnace at 550°C for more than 3 hours until white to grey ash was obtained, then crucible will be removed from furnace to a desiccator to cool and then will be weighed again. For the crude fiber, crude fat and crude protein contents, samples will be sent to SEAFDEC for analysis. Vitamin C, in the form of ascorbic acid will be determined by through Iodine titration Method, at CapSU-SRA Laboratory.

Selection of Panelist

This finished product will be evaluated as to color, chewiness, tenderness, and stickiness and overall acceptability by the evaluators, composed of 30 faculty/staff members and 15 students from Capiz State University- Pontevedra Campus.

Evaluation of the Product

The acceptability of Calamansi-Tapioca Pearls to CapSU Calamansi Juice will be evaluated by 15 evaluators per replication; there will be a total of 45 evaluators since the evaluation will be conducted in 3 trials. The treatments will be presented at random to the evaluators to avoid positional bias. The tasters will be composed of selected students, employees, and faculty of Capiz State University, Pontevedra Campus. The qualities of the product will be evaluated by the panelists with the use of scorecards for sensor evaluation. Testing of the product in each replication will be done. The evaluators will be oriented as to how to evaluate the product. They will be guided in determining the acceptability of the different product using score cards containing a rating scale in the different sensor qualities such as taste, appearance, color, aroma, and overall acceptability.

Data Analysis

The scores given by the evaluators will be analyzed using the arithmetic mean and standard deviation. Appropriate test of difference will be employed to the data set; analysis of variance (ANOVA) will be used if the data is normally distributed while Kruskal-Wallis H Test will be used if the data does not satisfy the assumption of normality. Significant differences among means will be further analyzed using an appropriate post-hoc test for the chosen statistical tool.

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(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.)

Activities	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
1. Gathering of materials						
2. Development of the Tapioca/Boba Pearls						
3. Incorporating Tapioca/Boba Pearls to Calamansi Juice						
4. Laboratory Analysis						
5. Sensory Evaluation						
6. Data Gathering						
7. Data Analysis						
8. Writing of Report						

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Publication

- at least 1 original article on the product development and testing presented/published
- at least 1 technical manual/guide
- at least 1 product specification

Patent/Intellectual Property

- at least 1 IP (Patent/UM) applications – method of preparation
- at least 1 trademark applied (product name and logo)
- at least 1 copyright applied (technical manual/guide)

Product

- at least 1 Zestful Infusion formulation
- at least 2 processes - process of preparing the tapioca pearls and process of incorporating the tapioca pearls to CapSU Calamansi Juice
- one (1) training program (on calamansi juice preparation)

People Service

- at least one (1) training conducted
- at least two (2) young faculty and research personnel trained
- at least one (1) BS student trained
- (1) Invention disclosure report
- (1) prior art search report
- (1) technology pricing report
- (1) IP valuation report
- (1) Cost Analysis report

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Place and Partnership

- 1 partnership with calamansi farmer forged

Policy

- at least one policy input on the healthy beverage production

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Calamansi producers
Product innovators
Researchers
Student Entrepreneurs

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
KRIS B. ROXAS	Project Leader	Food Chemistry
LEAH B. YAKIT	Project Staff	Food Science
CINDY MERISCO	Project Staff	Physical Science
CHRISTAL FE DOLLETE	Project Staff	English/Manuscript Writing

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Line Item Budget							
Particulars	Month						Total
	1	2	3	4	5	6	
I. PERSONAL SERVICES							
A. SALARIES AND WAGES							
1 research aide @ Php 10, 000 per month	11,068	11,068	11,068	11,068	11,068	11,068	66,408.00
Sub-total for PS	11,068.00	11,068.00	11,068.00	11,068.00	11,068.00	11,068.00	66,408.00
II. MAINTENANCE AND OTHER OPERATING EXPENSES							
Travelling Expenses	640.00	640.00	640.00	640.00	640.00	640.00	3,840.00
Office supplies	10,000.00						10,000.00
Ingredients		10,000.00					10,000.00
Other Professional Expenses (Lab Analysis)				30,000.00			30,000.00
Other Professional Expenses (IP Protection)					15,000.00		15,000.00
Sub-total for MOOE	10,640.00	10,640.00	640.00	30,640.00	15,640.00	640.00	68,840.00
III. EQUIPMENT OUTLAY							
Sub-total for EO							-
Grand Total	21,708.00	21,708.00	11,708.00	41,708.00	26,708.00	11,708.00	135,248.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Adubofuor, J., Amankwah, E. A., Arthur, B. S., & Appiah, F. (2010). Comparative study related to physico-chemical properties and sensory qualities of tomato juice and cocktail juice produced from oranges , tomatoes and carrots. African Journal of Food Science, 4(July), 427–433.

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)